Multi-Agent Price Discovery

Analytical HJB vs PPO HJB Market Makers in a Kyle-Style Informed Market

Team

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MARL Simulation Environment: Architecture

Core design:

- Rust-backed limit-order-book exchange (`_sim_exchange`)
- Python agent lifecycle: each agent implements `on_tick()`
- The loop is **agent-agnostic**: advances time, calls `on_tick` on every agent, collects nothing
- Agents self-log their own CSV metrics

Execution per tick:

1. Market ticks forward ($\Delta t = 1$)
2. Each agent's `on_tick()` runs and **blocks** until it returns
3. Market state updates from order executions
4. Repeat

```
for tick in range(n_ticks):
        market.tick()
    for agent in agents:
     agent.on_tick(sim_t)
        sim_t += 1
```

Agent Lifecycle & Information Flow

Each agent's `on_tick()` follows the same pattern:

1. **Check scheduler:** if it is not this agent's decision time, return immediately
2. **Drain acks/fills:** process confirmations and fills from the exchange since the last tick
3. **Read market data:** `md = market.get_md(asset_id)` – best bid/ask, full LOB, fill counters
4. **Build observation:** agent-specific function that maps raw market data to a fixed-dimension state vector
5. **Decide action:** sample from PPO policy (training) or deterministic forward pass (eval)
6. **Submit orders:** `action_to_orders()` produces limit/market orders, sent through the agent's connection
7. **Advance step counter:** increment internal decision step

All orders from all agents are collected by an `OrderShuffler`, interleaved randomly, then dispatched to the exchange in a single batch per tick.

Order Routing and Market Data

Order flow:

- Independent agents (HJB MM, PPO HJB MM, IT) each have their own conn to the exchange
- The `OrderShuffler` meta-agent wraps sub-agents (UniformMMs, Poisson noise) and collects orders from independent agents
- All orders are pooled into one list, `random.shuffle()` is applied, then each order is dispatched through its originating agent's conn
- This prevents any agent from consistently getting priority at the exchange

Market data returned by `get_md()`:

- `bid`, `ask` – best bid/ask prices
- `bids[N]`, `asks[N]` – LOB depth (price, volume)
- `taker_volume_buy/sell` – cumulative market order volume
- `informed` – direct informed-flow indicator (when available)
- Side-entry counters (nlab, nmb, nlas, nms)

PPO Training Lifecycle

Each trainable agent runs its own PPO loop independently:

Within a rollout:

- Collect n steps (default $n = 40$)
- At each step: observation $\rightarrow$ action $\rightarrow$ reward $\rightarrow$ next observation
- Store transition in rollout buffer: (s_t, a_t, r_t, s_{t+1})

At buffer full (every n steps):

1. Compute advantages via GAE(λ)
2. Run PPO update (K epochs, mini-batch SGD)
3. Clear buffer, increment episode counter
4. Call `on_episode_start()` to reset episode state

During the simulation:

- `total_timesteps` budget (default 20 000)
- Eval interleaving every `eval_freq` steps
- Checkpointing every `checkpoint_freq` steps
- Agents are completely independent – they do not share parameters, buffers, or rewards

What Are We Trying to Do?

Goal: Study price discovery in a limit-order-book market where:

- An **informed trader** (IT) knows the true fundamental value v and exploits it
- **Market makers** provide liquidity but face adverse selection risk
- We compare three MM configurations: (1) fixed analytical HJB MMs only, (2) learned PPO HJB MMs only, (3) both together
- IT is trained separately in each environment

Key question: Does a learned maker (PPO HJB) behave differently from an analytical one? Does coexistence change price discovery dynamics?

The Kyle-Style Informed Model

Classic Kyle (1985):

- Single informed trader, market makers, noise traders
- IT trades on private information, MMs set prices
- Equilibrium: MM learns λ (price impact), IT chooses order size

Our extension:

- Discrete-time tick-by-tick simulation ($\Delta t = 1$)
- IT action: $\beta \in [0, 10]$ linear policy
  $q = \beta \cdot (v - a_t)$
- IT reward: $(v - \text{fill price}) \cdot \text{volume}$
- IT is `buy_only` (only exploits positive mispricing)

Market Maker Models

Both MM types use the Guéant HJB framework:

- Solves a PDE backward on inventory grid $q \in [-20, 20]$
- Depth formula: $\delta_{\text{bid}} = \frac{1}{\kappa} + h(q) - h(q+1)$
- κ controls fill-probability decay per tick
- Running penalty ϕ and terminal penalty α on q^2

Analytical HJB MM (fixed)

- $\kappa = 0.8$, constant throughout
- $\lambda = 1.0$, fixed arrival rate
- No learning: quotes deterministically from PDE

PPO HJB MM (trainable)

- Learns $\kappa_t \in [0.01, 2.0]$ via PPO
- 7-dim observation: q, spread, λ est., mid vol, imbalance, t, κ_{t-1}
- Reward: $\Delta\text{cash} - \phi q^2 \Delta t - \alpha q^2 \cdot 1_{\text{last step}}$
- HJB PDE solved once at init with $\kappa_{\text{ref}} = 0.8$

The Market Environment

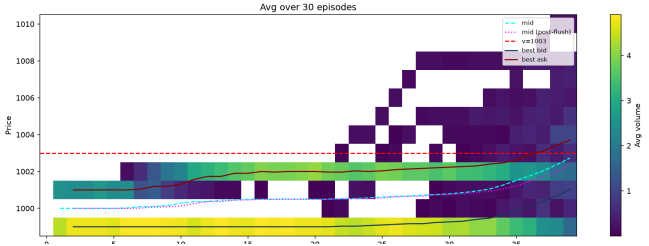
- Reference price: 1000, tick size: 1, lot size: 1
- Decision interval: $\Delta t = 1.0$, episode length: 40 steps
- Training: 20 000 total timesteps, 40 steps per rollout
- Evaluation: 30 independent episodes (seeds 10 000+ep)
- Rust-backed exchange (`_sim_exchange`), agents interleaved via `OrderShuffler`

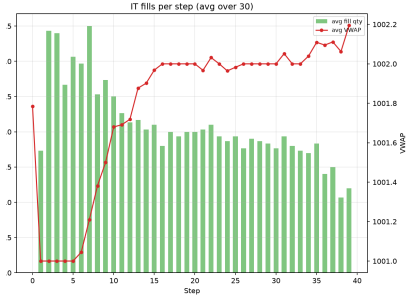
Agent	Role	Parameters
Poisson Noise	Random market orders	$\lambda^{\pm} = 1.0$
HJB MM ($\times 3$ or $\times 5$)	Fixed liquidity	$\kappa = 0.8$, $q \in [-20, 20]$
PPO HJB MM ($\times 3$ or $\times 5$)	Learned liquidity	$\kappa \in [0.01, 2.0]$, $q \in [-20, 20]$
Kyle Informed Trader	Directional flow	$\beta \in [0, 10]$, linear, buy-only

Six Experiments

Fair Value v	Configuration	HJB MMs	PPO HJB MMs
$v = 1003$ (3 ticks above ref)	HJB-only	5	0
	PPO-only	0	5
	Mixed (3+3)	3	3
$v = 1006$ (6 ticks above ref)	HJB-only	5	0
	PPO-only	0	5
	Mixed (3+3)	3	3

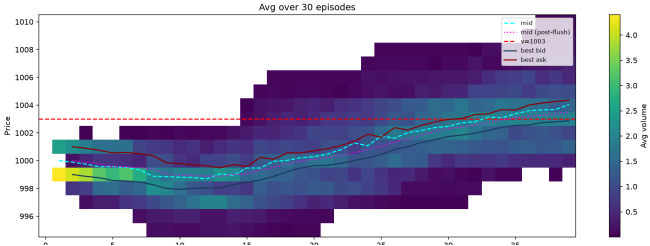
$v = 1003$ – HJB-Only (IT vs 5 analytical HJB MMs)

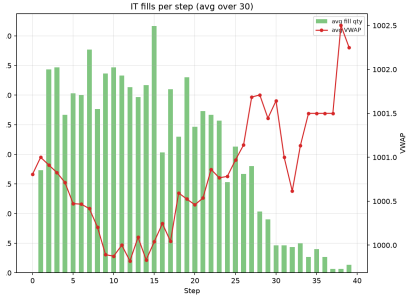




- Mid price converges smoothly toward $v = 1003$ (≈ 1002.5 by step 40)
- IT accumulates $\approx$ 20–25 lots; IT's β stays moderate ($\approx$ 2–3)
- MM PnL positive – they buy from noise, sell to IT at a mark-up

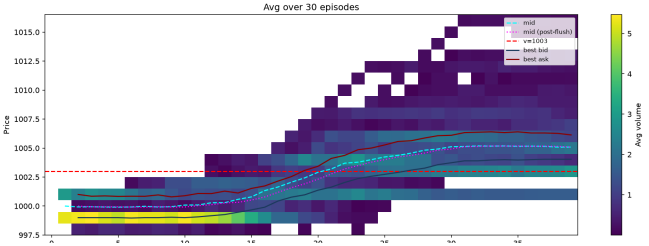
$v = 1003$ – PPO-Only (IT vs 5 PPO HJB MMs)

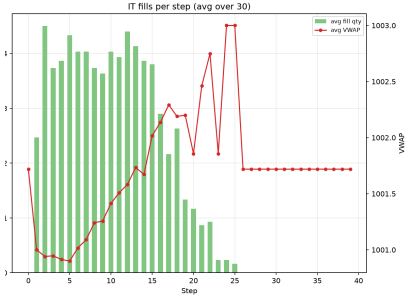




- **Dip-then-rise pattern**: mid drops to ~ 997 then climbs toward 1003
- κ starts ~ 1.15, dips to ~ 1.07, then rises to ~ 1.32
- PPO MMs widen spreads initially in response to informed flow, then tighten

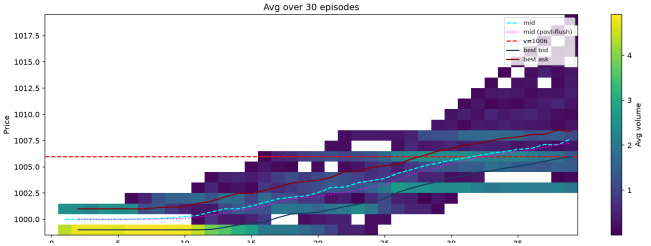
$v = 1003$ – Mixed (IT vs 3 HJB + 3 PPO HJB MMs)

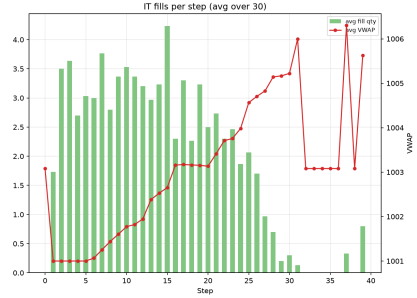




- Smooth convergence toward v – no dip, analytical HJB MMs anchor the mid
- PPO κ converges steadily to ~ 1.25–1.35 over the episode
- IT β oscillates $\sim$ 3–6 – higher than HJB-only due to more MM competition

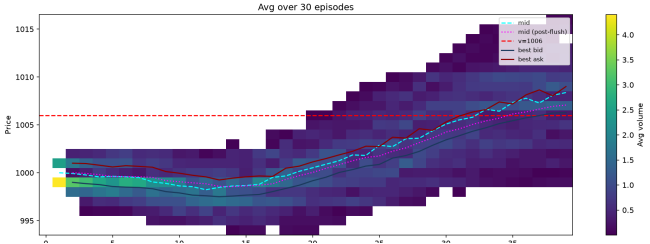
$v = 1006$ – HJB-Only (IT vs 5 analytical HJB MMs)

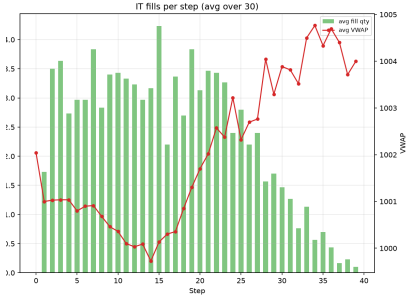




- Mid rises toward $v = 1006$, but ends ~ 1004 – incomplete convergence within 40 steps
- IT β oscillates $\sim$ 4–7, consistently higher than the $v = 1003$ case
- Larger fair value gap $\rightarrow$ larger mispricing $\rightarrow$ more aggressive IT

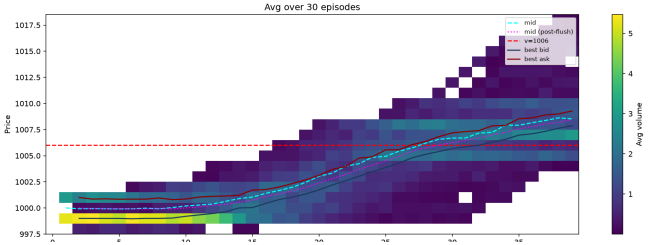
$v = 1006$ – PPO-Only (IT vs 5 PPO HJB MMs)

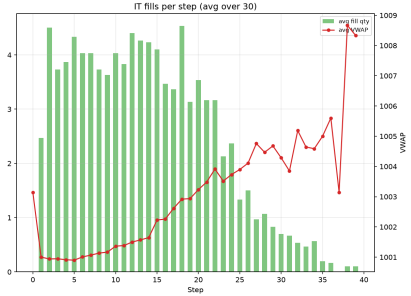




- Dip pattern observed but less pronounced than at $v = 1003$
- κ stays high (~ 1.2–1.4) – MMs less reactive because IT demand is already priced in
- Mid ends ~ 1004–1005, about 1–2 ticks below v

$v = 1006$ – Mixed (IT vs 3 HJB + 3 PPO HJB MMs)





- Stable convergence, same pattern as the $v = 1003$ mixed case
- PPO κ shows wider spread ($\sim$ 1.1–1.3) than the $v = 1003$ case
- IT $\beta \sim$ 4–8, the highest across all experiments (more MM targets = more trading)

HJB vs PPO: Key Differences

PPO-Only Behavior:

- **Dip-then-rise** signature: mid drops below ref_price before converging toward v
- κ drops initially (widening spreads) then rises (tightening)
- Without analytical anchor, MMs' collective quoting determines mid fully

Mixed Behavior:

- Smooth convergence – no dip
- Analytical HJB MMs ($\kappa = 0.8$) provide a fixed quoting baseline
- PPO MMs learn to quote slightly tighter ($\kappa > 1.0$) to compete
- IT can exploit both MM types – higher β, more execution

Metric	HJB-Only	PPO-Only	Mixed (3+3)
Mid convergence	Steady	Dip-then-rise	Smooth
κ range (PPO)	N/A	1.07–1.32	1.10–1.35
IT β (v=1003)	2–3	2–5	3–6
IT β (v=1006)	4–7	3–7	4–8

Effect of Fair Value v

$v = 1003$ **(3 ticks above ref)**

- Smaller mispricing signal
- IT moderates aggression ($\beta \sim 2$–5)
- Mid converges within ~ 0.5 ticks of v
- Clearer differentiation between MM types

$v = 1006$ **(6 ticks above ref)**

- Larger mispricing $\rightarrow$ more aggressive IT
- IT β reaches 7–8 in mixed case
- Mid ends 1–2 ticks below v – incomplete convergence within 40 steps
- MMs forced wider spreads to survive adverse selection

Takeaway: Larger fair value gaps create stronger adverse selection pressure.

The Dip-Then-Rise: Mechanism

PPO-only: mid drops below ref_price (~ 997) before rising toward v (~ 1002.5).

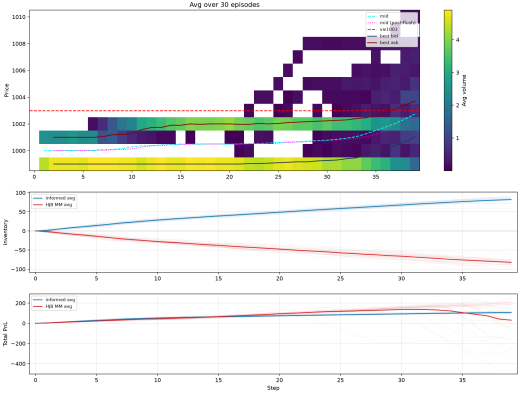
1. **Step 0:** PPO MMs start with $\kappa \approx 1.15$ (moderate spread). IT begins buying.
2. **Steps 1–10:** MMs observe fills, λ_{est} rises. They **lower** κ **to** ≈ 1.07 (widen spread) as protection.
3. **Asymmetric effect:** IT only hits the ask. No one hits the bid. $\rightarrow$ best bid drifts down, mid **drops**.
4. **Steps 10–30:** λ_{est} stabilizes, $h(q)$ inventory penalty already accounts for risk. MMs **raise** κ **to** ≈ 1.32.
5. **Steps 30–40:** Tighter quoting + IT buying drives mid toward v.

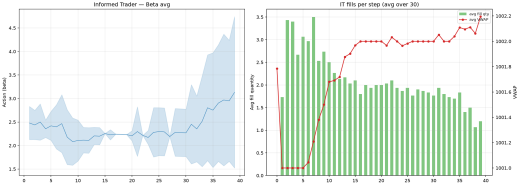
Why not in mixed? Analytical HJB MMs ($\kappa = 0.8$, symmetric) keep the bid anchored.

Summary

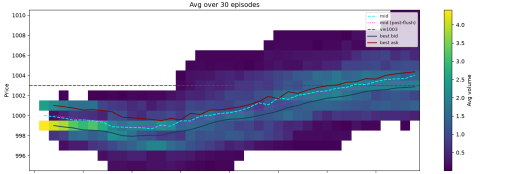
1. **PPO HJB MMs learn adaptive quoting** – they widen spreads in response to adverse selection and tighten once risk is priced in.
2. **The dip-then-rise pattern** emerges only in PPO-only environments where no analytical anchor exists.
3. **Mixed environments** produce the smoothest convergence. Analytical MMs provide a quoting floor; PPO MMs learn to compete by quoting slightly tighter.
4. **Fair value matters**: larger $v - p_0$ gaps amplify adverse selection, increasing IT aggression and slowing price discovery.
5. **3+3 MMs** were sufficient – earlier 5+5 experiments showed instability at $v = 1006$.

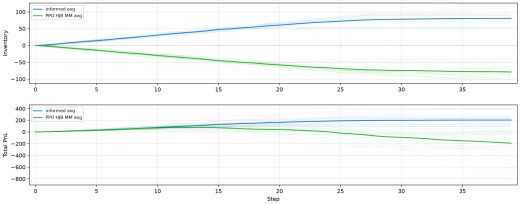
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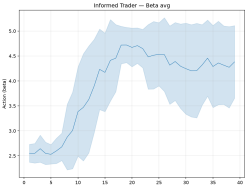


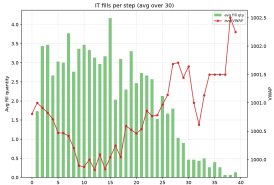


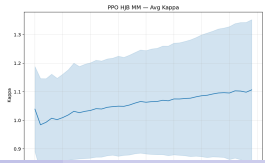
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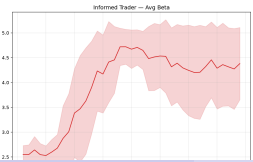




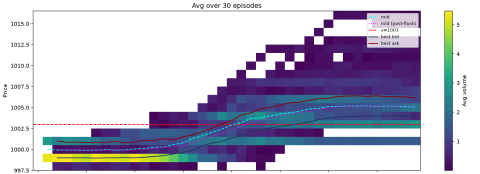


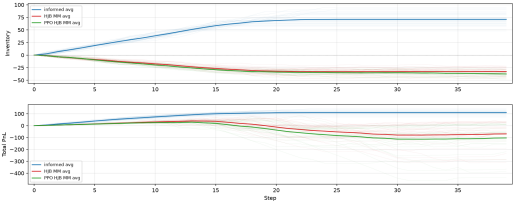


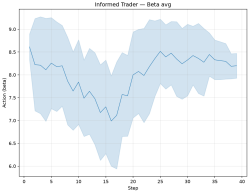


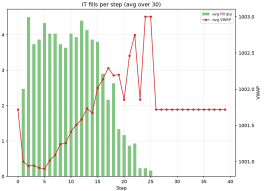


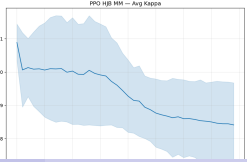
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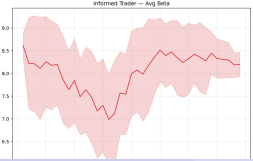




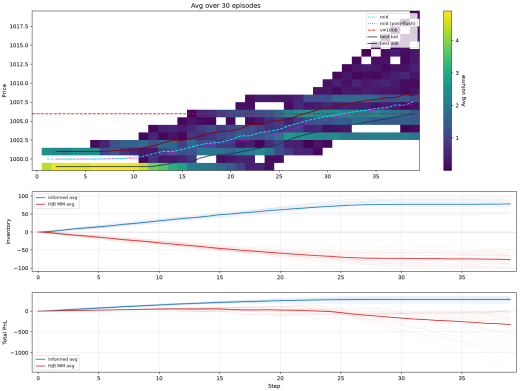


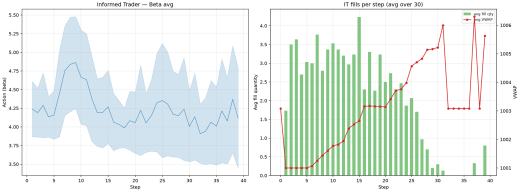




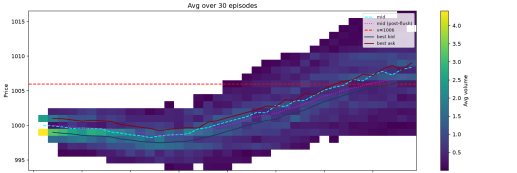


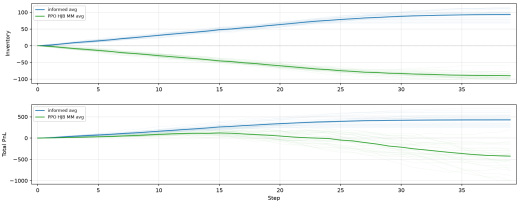
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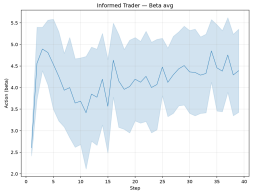


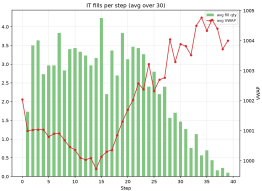


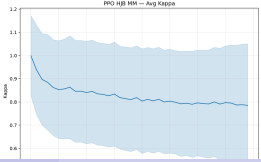
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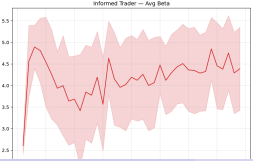












Annex: $v = 1006$ – Mixed (3+3)

